Geography Ch-18 (Must-Do): The Hot Desert & Mid-Latitude Desert Climate (BWh / BWk)

Tier 1: Must-Do | Source: GC Leong + Majid Husain (Indian & World Geography)
Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

Hot Deserts (Trade-Wind) & Mid-Latitude Deserts: Causes of Aridity, Extremes & Xerophytes

GS-I (Physical Geography - Climatology)
Geography

■ NEWS TRIGGER

Foundation concept (GC Leong Ch-18). Deserts are regions of scanty rainfall - HOT deserts of the Saharan type (Trade-Wind deserts) and TEMPERATE/mid-latitude deserts of the Gobi type (continental interiors). Aridity is the key-note.

■ INTRO & ORIGIN

Hot deserts lie on the western sides of continents around 15-30 deg N & S under the Sub-tropical High (descending, drying air) and offshore Trade Winds, often flanked by cold currents. Mid-latitude deserts lie deep in continental interiors, cut off from rain-bearing winds. Both are arid but differ sharply in temperature regime.

Hot-desert aridity: (1) descending air of the Sub-tropical High (Horse Latitudes); (2) offshore Trade Winds (dry by the time they reach the land); (3) cold offshore currents (chill air, suppress rain - e.g. Atacama/Namib). Mid-latitude aridity: interior location + rain-shadow of mountains (continentality).

■ KEY DATA

Feature	Hot Desert (BWh)	Mid-Latitude Desert (BWk)
Examples	Sahara, Thar, Arabian, Kalahari, Atacama	Gobi, Turkestan, Patagonia
Location	15-30 deg, W. margins, Sub-tropical High	Continental interiors, mid-latitudes
Cause of aridity	Offshore trades + cold currents + subsidence	Interior location + rain-shadow
Temperature	Very hot summers; mild winters	Hot summers BUT very cold winters
Diurnal range	Extreme (scorching day, cold night)	Extreme; large annual range too

■ STATIC THEORY

• **Three causes of hot-desert aridity (GC Leong):** offshore Trade Winds, the Sub-tropical High Pressure belt (Horse Latitudes, descending dry air), and cold ocean currents along western coasts (Benguela, Peru/Humboldt) which chill air and prevent rain.

• **Temperature:** hot deserts have the highest recorded shade temperatures on earth and an EXTREME diurnal range (very hot days, cold nights) because clear skies allow rapid daytime heating and night-time radiation loss. Mid-latitude deserts add a large ANNUAL range - very cold winters (Gobi below freezing).

• **Rainfall:** scanty (< 25 cm), erratic and unreliable; often convectional cloudbursts causing flash floods and wadis. Cold-current coastal deserts (Atacama, Namib) get fog/mist that nourishes thin vegetation.

• **Vegetation - xerophytic:** drought-resistant scrub - bulbous cacti, thorny bushes, long-rooted wiry grasses, scattered dwarf acacias; date palms where groundwater surfaces (oases). Adaptations: reduced/waxy leaves, thick stems (water storage), long roots, thorns.

• **Human life:** nomadic herders (Bedouin), oasis cultivators (date palms), and, increasingly, mineral (oil) and irrigation-based settlement.

■ MUST-KNOW FACTS

1. Hot deserts = 'Trade-Wind deserts', western margins, 15-30 deg, under the Sub-tropical High.

■■ UPSC TRAPS

✗ Deserts are always hot, including at night.

2. Three aridity causes: offshore trades, Sub-tropical High subsidence, cold ocean currents.	✓ Clear desert skies cause rapid night-time radiation loss, so nights are cold - an EXTREME diurnal temperature range.
3. Cold-current coastal deserts: Atacama (Peru current), Namib (Benguela current) - driest on earth.	✗ Cold ocean currents bring rain to coastal deserts.
4. Hot deserts = extreme DIURNAL range; mid-latitude deserts add extreme ANNUAL range (cold winters).	✓ Cold currents INCREASE aridity - they chill the air, form fog but suppress rainfall (Atacama, Namib are the driest).
5. Vegetation is xerophytic (cacti, thorn scrub, wiry grass); oases support date palms.	✗ Hot and mid-latitude deserts have the same temperature regime.
	✓ Mid-latitude (Gobi) deserts have very COLD winters (large annual range); hot deserts have mild winters.

■ MAINS ANGLE

Explain the multiple causes of desert aridity (subsidence, offshore trades, cold currents, continentality) and contrast the climate and vegetation of hot versus mid-latitude deserts.

■ STUDY LINK

Geography -> World Climatic Types -> Hot & Mid-Latitude Deserts (BW); links to Sub-tropical High & cold currents

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■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: UNCCD COP16 Riyadh 2024 - Global Fight Against Drought & Desertification

GS-III (Environment) + GS-II (International Institutions)
Geography - Current Affairs

■ NEWS TRIGGER

UNCCD COP16 (2-13 December 2024, Riyadh, Saudi Arabia) - the largest-ever UN land conference and first in the MENA region, marking the Convention's 30th year. It launched the Riyadh Global Drought Resilience Partnership (~USD 12+ billion pledged) but deferred a binding Global Drought Framework to COP17 (Mongolia, 2026).

■ INTRO & ORIGIN

As the world's drylands expand, the UN Convention to Combat Desertification (UNCCD) is the global response - and COP16 Riyadh is the applied, current face of the desert chapter's aridity and land-degradation themes.

UNCCD is one of the three 'Rio Conventions' (with UNFCCC-climate and CBD-biodiversity), mandated to combat Desertification, Land Degradation and Drought (DLDD) and achieve Land Degradation Neutrality (LDN) by 2030. India hosted COP14 (2019, New Delhi).

■ KEY DATA

Outcome	Detail	Note
Riyadh Drought Resilience Partnership	~USD 12+ billion pledged	Crisis response -> long-term resilience
Global Drought Framework	Negotiated, NOT finalised	Deferred to COP17 Mongolia (2026)
Great Green Wall support	New funding (Italy, Austria)	Sahel restoration, 22 countries
VACS (crops & soils)	~USD 70 million	Climate-resilient food systems

■ STATIC THEORY

• **UNCCD:** one of the three Rio Conventions (1994); goal = combat DLDD and reach Land Degradation Neutrality by 2030 (SDG 15.3). India is a Party and hosted COP14 (New Delhi, 2019).

• **COP16 firsts:** largest UN land conference; first UNCCD COP in the Middle East (Riyadh); focused squarely on DROUGHT RESILIENCE and land restoration finance.

• **Key gap:** a legally binding global drought regime was NOT agreed - pushed to COP17 in Mongolia (2026); private finance remains small (~6%).

■ MUST-KNOW FACTS

1. UNCCD COP16: Riyadh, Saudi Arabia, 2-13 December 2024 - largest UN land conference, first in MENA.

2. Riyadh Global Drought Resilience Partnership: ~USD 12+ billion pledged.

3. Binding Global Drought Framework deferred to COP17 (Mongolia, 2026).

4. UNCCD = one of three Rio Conventions; India hosted COP14 (New Delhi, 2019).

5. Goal: Land Degradation Neutrality by 2030 (SDG 15.3).

■■ UPSC TRAPS

✗ COP16 adopted a binding global drought treaty.

✓ It launched a voluntary resilience PARTNERSHIP and pledges; the binding framework was deferred to COP17 (Mongolia, 2026).

✗ UNCCD deals with climate change mitigation like the UNFCCC.

✓ UNCCD specifically addresses desertification, land degradation and drought (DLDD) - a distinct Rio Convention from the UNFCCC.

■ MAINS ANGLE

Discuss UNCCD COP16 (Riyadh) as the global institutional response to expanding drylands, and evaluate why binding drought governance remains elusive.

■ STUDY LINK

Geography -> Deserts/aridity + GS-III land degradation (UNCCD, LDN, Rio Conventions); India hosted COP14

End of Register Notes — UPSC Agent / Copilot CLI